DS UA 9201 - Causal inference
Observational data - Confounding

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Previously, on the causal inference course...

We focused on **randomized experiments**:

- How they follow the crucial assumptions necessary for causal inference.
- What estimators of the ATE have good properties in randomized experiments.

We also discovered:

- **Randomization inference**, a type of statistical inference that relies **only** on the experimental design and no other assumption.
- **Covariates** can identify heterogeneous treatment effects.
- Methods to perform statistical inference for the CATE.
- Covariates can reduce variance by stratification.
- Post-treatment covariates and why to avoid them.
- The dangers of multiple hypothesis testing.

From now on: observational design

What to do when you cannot run an experiment?

- You find an available dataset.
- You don't get to randomly assign the treatment $\rightarrow$ You need to make assumptions about the observed treatment assignment process.
- Covariates become essential to characterize the causal relationship between T and Y.

This is an observational study.

Outline

1. Observational designs: overview
2. Natural experiments
3. Confounding
4. Conditional ignorability and positivity

Why can't we always conduct a randomized experiment?

- 1. They can be unethical (e.g. effect of smoking on lung cancer).
- 2. Costs may be high:
  - Need for a large sample size to assess treatment effect heterogeneity. Hence expensive (in terms of money, human resources, time...).
  - Difficult to recruit people.
- 3. Bias in population selection:
  - The whole population may not be represented in the randomized experiment.
  - For instance, in medical studies, old patients, or patients with comorbidities are less represented.
- 4. Hard to follow people over time:
  - Clinical trials may lack follow-up: tedious and expensive to follow everyone for many years.
  - The treatment effect is often computed after a pre-defined time point only (*e.g.* 1 year, 5 years).
  - And do not enable to search for long-term effects.

Lots of observational data available

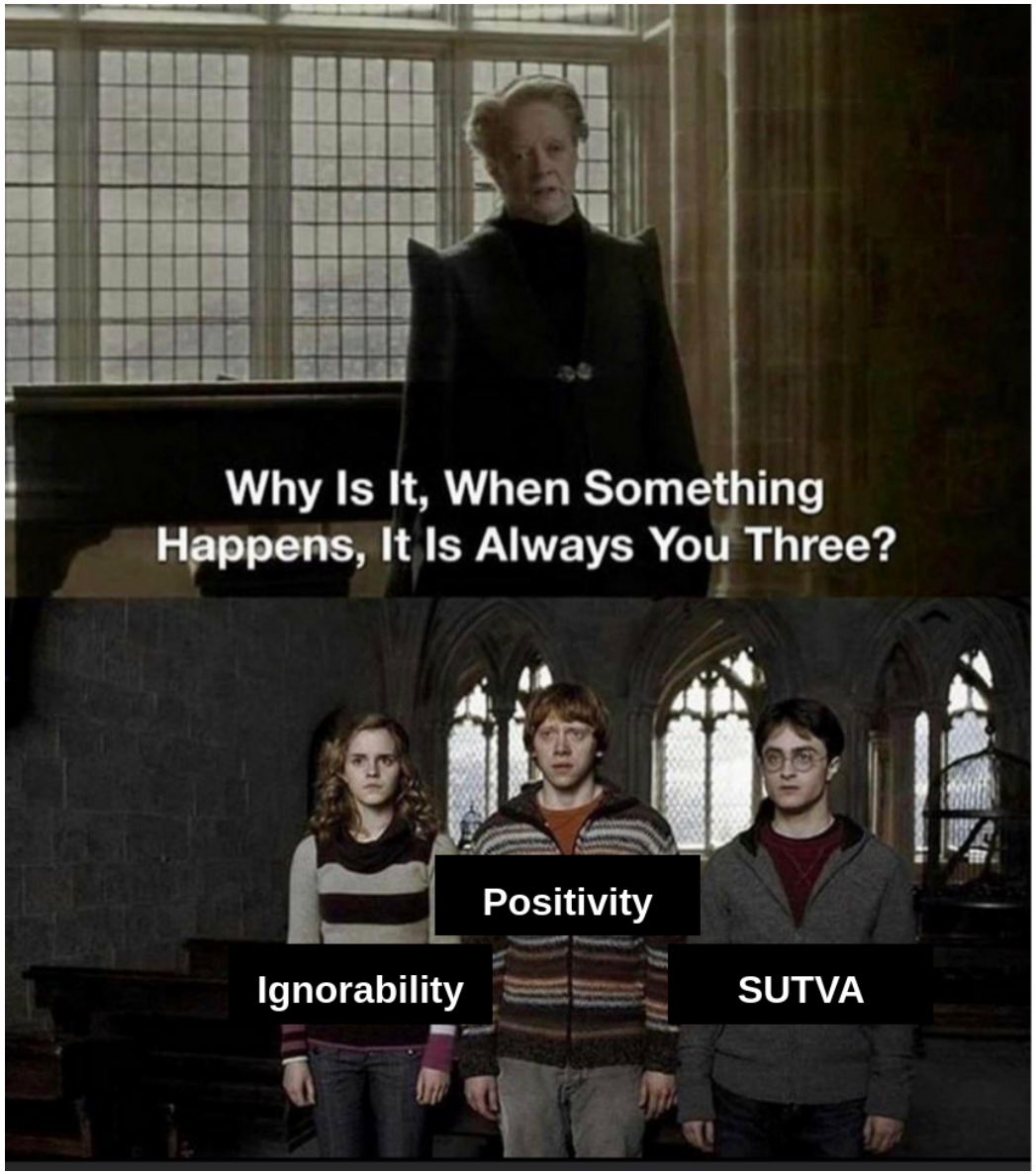
- There is a lot of data available, not necessarily collected to answer the causal question.
  - Electronic health records, medical health insurance data.
  - Surveys, offline or online.
  - Smart watches for number of steps, etc.
  - Social media data.

For example, we investigate whether the intake of anti-diabetic drugs increase the risk of breast cancer.

- Option 1: Run a randomized clinical trial by selecting a number of units, and assign half of them to take the anti-diabetic drug each day during one year.
- Option 2: Look at health insurance claims data, using which we identify both the delivery of anti-diabetic drugs; and the occurrence of breast cancer.

Discussion for next weeks:
Can we use this data to draw some causal conclusions?
If yes, under which assumptions? And using what methods?

Reminder: identification of the ATE



Reminder: identification the ATE

In order to identify the ATE using the difference-in-means estimator we need:

1. Positivity: $0 < \Pr(T_i = t) < 1$ for all treatment levels, t.
2. Ignorability: $Y_i(1), Y_i(0) \perp\!\!\!\perp T_i$, i.e., treatment is assigned independently of the potential outcomes.
3. SUTVA (Stable Unit Treatment Values Assumption): $Y_i = Y_i(1)T_i + Y_i(0)(1 - T_i)$, i.e., the observed outcome is either one of the potential outcomes of interest.

In a randomized experiment, we directly construct $\Pr(T_i = t)$ so that 1 and 2 are satisfied.

Are they verified in observational data?

In this case, you do not control directly treatment assignment.

The Problem

Can you guarantee that positivity and ignorability hold?

- What is $\Pr(T_i = t)$ in this case?
- We will use what we know about $\Pr(T_i = t)$ and make assumptions to make sure they hold.

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The simplest case: A Natural Experiment

A natural experiment is a setting in which[1]:

- The probability of treatment assignment is not directly controlled by the researchers (not a randomized experiment).
- The distribution of treatment assignment is unknown to the researchers (we do not know the exact value of $\Pr(T_i = t)$).
- But the treatment is still randomly assigned due to external events.

Ignorability

If you have a natural experiment - you know the treatment was assigned independently of the potential outcomes - then you can just treat it as a randomized experiment (provided that SUTVA is satisfied).

[1] Cataneo et al, "A Practical Introduction to Regression Discontinuity Designs"; Cambridge Elements: Quantitative and Computational Methods for Social Science, 2024.

The simplest case: A Natural Experiment

How confident are you that the treatment was randomly assigned?
This information is metadata, it cannot be assessed *a posteriori* with the data.
Here are some examples:

- Weather and voter turnout[2]
- Lockdown for covid and reduction in fetal mortality[3]

But in practice: rare, often based on unexpected events.

[2] "The Republicans Should Pray for Rain: Weather, Turnout, and Voting in U.S. Presidential Elections"; Brad T. Gomez, Thomas G. Hansford, and George A. Krause, The Journal of Politics 2007

[3] "Unprecedented reduction in births of very low birthweight and extremely low birthweight infants during the COVID-19 lockdown in Ireland: a 'natural experiment' allowing analysis of data from the prior two decades"; Philip RK et al., BMJ Glob Health. 2020

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In general, there are confounders

- In the vast majority of observational designs, ignorability will not hold.
- And this is due to the presence of **confounders**.[4]

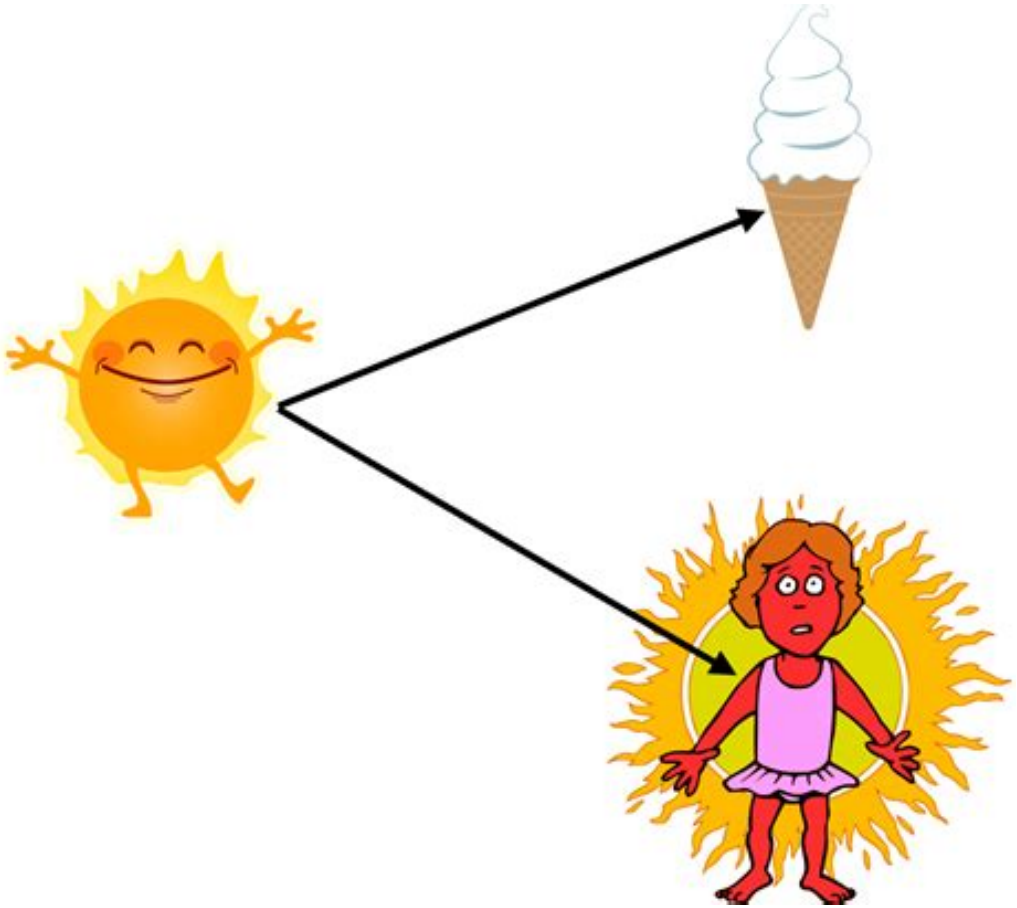
Confounder

A confounder is a common cause of both the treatment and the outcome.

- Generally, we denote confounders by X.
- We might think the treatment is causing the outcome, but there might be a hidden factor that is causing both.
- Confounding induces bias in the estimation of the ATE.
- Confounders creates “spurious correlation" between observed treatment and observed outcome.
- Confounders are **pre-exposure** covariates!

[4] We say there are **confounding** effects.

Examples of confounders 1



Spurious correlation!

Example of confounders 2

Does the intake of anti-diabetic drugs increase the risk of Breast Cancer?

$$\begin{tikzcd} X \arrow[d] \arrow[dr] & \\ T \arrow[r] & Y \end{tikzcd}$$

- T: Intake of anti-diabetic drugs
- Y: Occurrence of Breast Cancer
- X: Obesity

Obesity as a confounder:

- Obesity (X) increases the risk of breast cancer (Y).
- Obesity (X) is linked with diabetes; with the use of anti-diabetic drugs (T).

Examples of confounders 3

Another Benefit to Going to Museums? You May Live Longer

Researchers in Britain found that people who go to museums, the theater and the opera were less likely to die in the study period than those who didn't.

Examples of confounders 4

Mathematical definition for confounding

Slightly more technically, a confounder X_i:

- Is associated with the probability of treatment assignment: $\Pr(T_i = 1|X_i = 1) \neq \Pr(T_i = 1|X_i = 0)$
- Is associated with the potential outcomes: $E[Y_i(t)|X_i = 1] \neq E[Y_i(t)|X_i = 0]$ for $t = \{0, 1\}$

$$\begin{tikzcd} X \arrow[d] \arrow[dr] & \\ T \arrow[r] & Y \end{tikzcd}$$

- It implies that T is associated with the potential outcomes.
- Therefore, ignorability does not hold.

What about positivity?

- Positivity is not guaranteed anymore in observational designs.
- Example: if T is the intake of anti-diabetic drugs, then treated patients must be diabetic.
- Non-diabetic patients i are such that $Pr(T_i = 1) = 0$.
- So this assumption should also be verified before an observational analysis.

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A solution

- Even when strong ignorability and positivity do not hold, they may hold conditionally on a certain set of confounding variables.
- If this set is well defined, we will be able to estimate causal effects.
- We will need different causal estimands than the difference-in-means.
- Let's first define more precisely our new causal assumptions: **conditional positivity** and **conditional ignorability**.

Conditional Positivity

We assume that positivity holds conditionally on the covariates:

Definition

We assume that $0 < \Pr(T_i = 1 | X_i = x) < 1$, for all x in the domain of X and treatment levels t.

Treatment is non-deterministic in the covariates:

- Knowing a unit's covariate value will never determine certainly what treatment that unit gets.
- Usually randomness comes from other factors that are not related to the outcome.

Left: Non smoker and never treated Right: Smokers and all treated

Conditional Ignorability

We also assume that the treatment is independent of the potential outcomes conditional on the covariates:

Definition

We assume that $Y_i(1), Y_i(0) \perp\!\!\!\perp T_i | X_i = x$ for all x and t.

- Covariates help "reveal the true story" behind the treatment assignment process.
- There are no omitted factors that could be causing both the treatment and the outcome
- Synonyms: no unmeasured confounding, selection on observables, no omitted variables, exogeneity, conditionally exchangeable, etc...

Consequence of conditional ignorability

Important note:

- We cannot write:

$$E[Y_i(1)|T_i = 1] = E[Y_i(1)]$$

- However, conditional ignorability allows us to write:

$$E[Y_i(1)|T_i = 1, X_i = x] = E[Y_i(1)|X_i = x]$$

- Within levels of X_i, treatment is assigned “as-if-random".

New identification strategy: selection on observables

Selection on observables

In a selection-on-observables identification strategy, we assume that treatment assignment is ignorable conditional on observed covariates $\mathbf{X}$. (Obviously, we still assume SUTVA!)

We assume that we have all the information we need to characterize the causal relationship between the outcome and treatment assignment.

Causal estimand	Causal assumptions	Statistical estimand	Statistical estimator
?	SUTVA Conditional positivity, Conditional ignorability	?	?

Does this remind you of anything? Under our selection-on-observables design, we can easily identify and estimate CATE.

Next in the causal inference course

- It is time to introduce directed causal graphs.
- We have naturally used graphs with nodes and arrows to represent causal effects between variables.
- But we never properly defined them, and their relationship with causal assumptions.
- They will become very useful to find the (minimum) set of covariates for which conditional ignorability holds.
- Later, we will discover several identification strategies and causal estimators for observational designs (stratification, inverse probability weighting, matching, etc...).
- And we will discuss why the classical difference-in-means fails here.